

AAYUSH GUPTA

Senior Gen AI Engineer | Multi-Agent Systems & RAG Architecture | Production LLM & Voice AI

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PROFESSIONAL SUMMARY

Gen AI Engineer with **6+ years** of experience building production-grade Generative AI, Machine Learning, and Deep Learning systems. Deep hands-on expertise in **LLMs (Claude, OpenAI GPT, Gemini)**, **RAG architecture**, **multi-agent systems (LangChain, LangGraph)**, **vector databases (Pinecone, FAISS)**, and **prompt engineering**. Currently architecting a production conversational AI platform: a multilingual Voice AI virtual assistant with 103 tools across 9 MCP servers, plus an LLM-powered analytics dashboard with Pinecone-backed semantic retrieval. Strong MLOps/LLMOps foundation — LLM-as-judge evaluation, LangSmith/Langfuse tracing, CI/CD, and scalable FastAPI deployments on AWS Bedrock, Azure OpenAI, and GCP.

TECHNICAL SKILLS

Generative AI & LLMs: OpenAI (GPT-4), Claude (Anthropic), Gemini, RAG Architecture, Prompt Engineering, Agentic Workflows, AI Agents & Multi-Agent Systems, LangChain, LangGraph, MCP (Model Context Protocol), Conversational AI, Voice AI (STT/TTS, Hinglish multilingual), Document Intelligence (NLP over unstructured text)

Vector Databases & Retrieval: Pinecone, FAISS, semantic search, hybrid retrieval, embedding pipelines (Sentence Transformers, bge-m3), Neo4j knowledge graphs

ML / Deep Learning: Machine Learning, Deep Learning, NLP, Transformers, BERT, Computer Vision, PyTorch, scikit-learn, Spacy, NLTK

MLOps & LLMOps: LLM-as-judge evaluation frameworks, LangSmith, Langfuse, prompt/version tracing, evaluation pipelines, drift detection, MLflow, structured logging, observability & monitoring

Backend & Deployment: Python (Advanced), FastAPI, SQL, REST API integrations, WebSocket streaming, Docker, scalable concurrent deployments

Cloud & CI/CD: AWS Bedrock, Azure OpenAI, Azure ML, Azure Databricks, GCP, Terraform (IaC), GitHub Actions, GitLab CI, Jenkins

Databases: PostgreSQL, Neo4j, Redis, vector stores

PROFESSIONAL EXPERIENCE

AI Engineer | Quara Digital — January 2024 – Present

- **Production Conversational AI — Multi-Agent Voice Assistant:** Architected and deployed a production agentic system for multilingual (Hinglish + English) voice interactions covering property discovery, lead qualification, and workflow automation. Multi-layer architecture: NestJS gateway → FastAPI orchestration engine → 9 domain-specific MCP servers exposing 103 tools through a Claude-based tool-calling loop. Real-time STT/TTS pipelines over WebSocket streaming with sub-300ms voice response latency; context compression delivering 40% token cost reduction.
- **AI Analytics Dashboard — RAG over Enterprise Data:** Built an LLM-powered analytics system translating natural language to optimized SQL over a 1,200+ table schema, returning human-readable insights and interactive visualizations. Implemented **Pinecone**-backed semantic search for schema and catalog retrieval, with a user-feedback loop that refines prompts for continuous improvement.
- **LLM-as-Judge Evaluation Framework:** Designed automated response-quality scoring with a dedicated evaluator LLM across accuracy, tone, completeness, and domain compliance. Wired scores into a continuous feedback loop for prompt optimization, cutting manual QA overhead by 60%. Full observability: structured logging of tool invocations, exponential-backoff retries, real-time alerting on score degradation.
- **Property Valuation & Forecasting API (Classical ML):** Production regression system on FastAPI for real-time appraisals from geospatial and structural features; drift-detection pipeline with automated retraining and A/B-tested rollouts on Azure ML. Reduced manual valuation time by 80%.
- **Buyer Clustering & Computer Vision Pipelines:** Unsupervised segmentation of high-net-worth leads driving a 35% conversion improvement; multithreaded room-type classification processing 50k+ property images monthly at sub-200ms inference latency.
- **CI/CD & Infrastructure:** GitHub Actions, GitLab CI, and Jenkins pipelines; Terraform IaC for Azure provisioning; monitoring dashboards, structured error tracking, zero-downtime deployments.

Lead Data Scientist | RapiPay Fintech — March 2023 – December 2023

- **Document Intelligence for Risk Assessment:** NLP pipelines over unstructured SMS data and CIBIL credit reports to flag defaulters and fraudulent loan applications — 92% fraud-detection accuracy, 40% reduction in manual review. Containerized FastAPI ML microservices on AWS with auto-scaling.
- **Model Lifecycle Management:** End-to-end lifecycle from exploration to production monitoring with real-time high-risk alerts; generator-based feature engineering reducing memory overhead by 30% on large-scale financial datasets.

Data Scientist | Tiger Analytics — January 2022 – March 2023

- **Call Center Quality Audit Automation:** Fine-tuned Transformer models (BERT, Hugging Face) for call authentication, phase-level sentiment analysis, transfer-point identification, and intent extraction — 95% accuracy. Stakeholder dashboards in Streamlit.
- **Aspect-Based Sentiment Analysis:** Custom Spacy NER + PyABSA processing 100k+ reviews monthly on Azure Databricks distributed compute over large audio-to-text datasets.

Software Engineer & Data Scientist | Mindtree — September 2019 – December 2021

- **Healthcare / Pharmaceutical Market Intelligence:** Automated Python web-scraping pipelines aggregating 50+ sources into a structured data foundation supporting large-scale pharmaceutical investment decisions.
- **5G Data Optimization:** Optimized 5G data-streaming protocols for Tier-1 telecom clients; refactored legacy codebases with OOP principles for maintainability and scale.

EDUCATION

B.Tech. in Electronics & Telecommunication — Guru Gobind Singh Indraprastha University (GGSIPU), 2019

CERTIFICATIONS

Applied Machine Learning Course